

The language share, step by step

Waiz Khan — August 2026

1. The data

Take N sentences and their translations into L languages, so that sentence c carries the same meaning in every language. Feed each of the $L \times N$ strings to the model and mean pool the hidden states at one layer. This gives one vector per cell of an $L \times N$ grid:

$$h_{\ell c} \in \mathbb{R}^D, \quad \ell = 1, \dots, L \text{ (languages)}, \quad c = 1, \dots, N \text{ (sentences)}.$$

In the runs reported here $N = 300$, $L = 41$, and D is the model's hidden size. Everything below is done **separately for each coordinate d** of the vector and then summed over d ; the ratio at the end is unchanged by whether one sums or averages over coordinates, so the choice does not matter.

2. The model being fitted

For coordinate d , write the value in cell (ℓ, c) as

$$h_{\ell c d} = \mu_d + \underbrace{a_{cd}}_{\text{sentence}} + \underbrace{b_{\ell d}}_{\text{language}} + e_{\ell c d}, \quad a_{cd} \sim (0, \sigma_{C,d}^2), \quad b_{\ell d} \sim (0, \sigma_{L,d}^2), \quad e_{\ell c d} \sim (0, \sigma_{E,d}^2),$$

all mutually independent. This is the balanced two way crossed random effects layout with one observation per cell. In words: a vector's position is the overall mean, plus an offset shared by every translation of that sentence, plus an offset shared by every sentence in that language, plus what is left.

One caveat, stated up front. With a single observation per cell, $e_{\ell c d}$ absorbs both measurement noise and any genuine language by sentence interaction; the two cannot be separated without replicates. So σ_E^2 should be read as “everything not explained by an additive sentence effect and an additive language effect”.

3. Sums of squares

Let

$$\bar{h}_{\ell \cdot d} = \frac{1}{N} \sum_c h_{\ell c d}, \quad \bar{h}_{\cdot c d} = \frac{1}{L} \sum_\ell h_{\ell c d}, \quad \bar{h}_{\cdot \cdot d} = \frac{1}{LN} \sum_{\ell, c} h_{\ell c d}.$$

Then

$$\begin{aligned} \text{SS}_{L,d} &= N \sum_{\ell=1}^L (\bar{h}_{\ell \cdot d} - \bar{h}_{\cdot \cdot d})^2, & \text{SS}_{C,d} &= L \sum_{c=1}^N (\bar{h}_{\cdot c d} - \bar{h}_{\cdot \cdot d})^2, \\ \text{SS}_{E,d} &= \sum_{\ell, c} (h_{\ell c d} - \bar{h}_{\ell \cdot d} - \bar{h}_{\cdot c d} + \bar{h}_{\cdot \cdot d})^2, \end{aligned}$$

with degrees of freedom $L - 1$, $N - 1$ and $(L - 1)(N - 1)$. Mean squares are $\text{MS} = \text{SS}/\text{df}$.

4. From mean squares to variances

For this layout the expected mean squares are

$$\mathbb{E}[\text{MS}_{L,d}] = \sigma_{E,d}^2 + N \sigma_{L,d}^2, \quad \mathbb{E}[\text{MS}_{C,d}] = \sigma_{E,d}^2 + L \sigma_{C,d}^2, \quad \mathbb{E}[\text{MS}_E] = \sigma_{E,d}^2.$$

Solving gives the estimates

$$\hat{\sigma}_{E,d}^2 = \text{MS}_{E,d}, \quad \hat{\sigma}_{L,d}^2 = \frac{\text{MS}_{L,d} - \text{MS}_{E,d}}{N}, \quad \hat{\sigma}_{C,d}^2 = \frac{\text{MS}_{C,d} - \text{MS}_{E,d}}{L}$$

The subtraction of MS_E is the noise correction: a language effect must exceed what the residual alone would produce before it counts as signal. These can come out negative when the true component is near zero; they are kept as computed in the moment estimator and counted, and the primary estimate uses nonnegative restricted maximum likelihood, which for this balanced design has a closed form: it agrees with the above when both are positive and sets the offending component to zero otherwise, re-pooling its sum of squares into the residual.

5. The quantity

Summing over coordinates,

$$\sigma_L^2 = \sum_d \hat{\sigma}_{L,d}^2, \quad \sigma_C^2 = \sum_d \hat{\sigma}_{C,d}^2, \quad \boxed{\text{language share} = \frac{\sigma_L^2}{\sigma_L^2 + \sigma_C^2}}$$

It is 0 when language contributes nothing (all L translations of a sentence sit at the same point) and 1 when meaning contributes nothing (all N sentences in a language sit at the same point). Measured across 33 models it runs from 0.59 to 0.90.

6. A worked example

One coordinate, $L = 2$, $N = 3$:

	c_1	c_2	c_3	row mean
English	1.0	3.0	5.0	3.0
German	2.0	4.0	6.0	4.0
column mean	1.5	3.5	5.5	3.5 (grand)

$\text{SS}_L = 3[(3 - 3.5)^2 + (4 - 3.5)^2] = 1.5$, so $\text{MS}_L = 1.5/1 = 1.5$.

$\text{SS}_C = 2[(1.5 - 3.5)^2 + 0 + (5.5 - 3.5)^2] = 16$, so $\text{MS}_C = 16/2 = 8$.

Every residual $h_{\ell c} - \bar{h}_{\ell} - \bar{h}_{\cdot c} + \bar{h}_{\cdot\cdot}$ is zero here, so $\text{MS}_E = 0$.

Hence $\hat{\sigma}_L^2 = (1.5 - 0)/3 = 0.5$ and $\hat{\sigma}_C^2 = (8 - 0)/2 = 4$, giving a language share of $0.5/4.5 = \mathbf{0.11}$: meaning dominates.

7. Why the ratio cannot see a uniform shrinkage

Suppose the model's representations are scaled by a constant, $h_{\ell c} \mapsto \kappa h_{\ell c}$. Every sum of squares scales by κ^2 , so every mean square does, so every variance component does:

$$\hat{\sigma}_L^2 \mapsto \kappa^2 \hat{\sigma}_L^2, \quad \hat{\sigma}_C^2 \mapsto \kappa^2 \hat{\sigma}_C^2 \quad \implies \quad \frac{\kappa^2 \sigma_L^2}{\kappa^2 \sigma_L^2 + \kappa^2 \sigma_C^2} = \frac{\sigma_L^2}{\sigma_L^2 + \sigma_C^2}.$$

The share is exactly invariant. In the worked example above, replacing every cell by one tenth of its value leaves the answer at 0.11 while σ_C^2 falls from 4 to 0.04.

This is not an estimation problem, and no better estimator of the same ratio avoids it: any statistic that is homogeneous of degree zero in the components is blind to a common rescaling. The consequence observed in the training study is that a checkpoint which had lost about 95 percent of σ_C^2 scored best in the study on this quantity.

The remedy is to report σ_C^2 itself alongside the ratio, which is what the collapse reading does: measured against a reference checkpoint it recovers the injected shrinkage exactly, returning $(1 - m)^2$ to within 0.01 for a shrinkage of magnitude m .

8. What is done with it

The share is one of four readings. The others do not use this decomposition at all: sentence matching asks, for each sentence in language ℓ , whether its English counterpart is the nearest of the N candidates; the tail reading restricts that to the worst aligned languages; the collapse reading is σ_C^2 against a reference. Each is converted to a percentile against a fixed grid of 33 models before any averaging, since the four are on different scales.